

SNIFFING LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
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Submission Platform:	Canvas
Report Date:	2026-01-07
Report Version:	1.0
Lab Title:	Sniffing
Module Reference:	Sniffing
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-01-07
Time Spent:	Approximately 1 hour
ENVIRONMENT	
Platforms and Systems:	
- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox. - Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Server), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Workstation), and PLABKALI01 (Kali Linux 2019.2 - Linux Kali Workstation). - The work in this module centered primarily on PLABWIN10 for MAC spoofing, packet capture, traffic inspection, and ARP-sniffing prevention review, with the broader domain environment present in the lab topology. - Remote access to the virtual machines was provided through the course platform.	
Context and Constraints:	
- The work performed in this environment covered MAC addressing, MAC spoofing tools, Wireshark, Sniff-O-Matic, and XArp for sniffing awareness and prevention analysis. - Windows Updates were disabled in the Practice Labs environment, so some software behavior reflected a guided legacy setup rather than a current hardened baseline. - The module aligned most closely with exam objectives 3.2 (Information Security Attack Detection) and 3.3 (Information Security Attack Prevention). - The overall workflow focused on how sniffing can be enabled, observed, analyzed, and reduced through better visibility and prevention-oriented tooling.	
Authorization and Scope:	
- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes. - No public IP space, real organizations, or unauthorized targets were involved. - No non-lab credentials or real-world data were collected, stored, or reused. - The actions documented here were performed for defensive understanding, network visibility, and ethical hacking education within the course environment. - Restricted material and lab credentials are omitted or redacted from this report.	
Work Objectives:	
- Examine how MAC address manipulation can affect host identity at the network interface layer. - Use Wireshark and Sniff-O-Matic to observe and analyze live network traffic. - Use multiple sniffing-related tools to compare how connections, packets, and identifiers are presented for analysis. - Review preventive controls through XArp and broader anti-sniffing practices such as protocol hardening, switching, encryption, and ARP-based monitoring.	

TECHNICAL ANALYSIS

MAC Address Manipulation with Change MAC Address:

- On PLABWIN10, Internet Explorer was opened and the internal Tools > Hacking Tools area was used to download and install Change MAC Address.
- The installation was completed with default options, after which the application launched in evaluation mode.
- From the Change MAC address pane, multiple MAC-generation modes were exercised, including Fully random, Random manufacturer from list, Only random device ID, and Random manufacturer from list with random device ID.
- Each selection altered the displayed MAC address, and in the manufacturer-based options, the vendor field changed as well.
- After applying the change, the network connection was restarted and the updated MAC address was refreshed and verified within the interface.
- The MAC lookup feature was also opened to inspect vendor identification behavior for a given address.
- This stage showed how easily layer-2 identity can be altered from userland tooling, which has direct implications for spoofing, attribution, and local network trust assumptions.

MAC Spoofing with SMAC 2.0:

- Still on PLABWIN10, SMAC 2.0 was downloaded from the same Hacking Tools area and installed using the default wizard selections.
- After accepting the license agreement and continuing in evaluation mode, the application displayed the MAC addresses of the available NICs in the system.
- A selected NIC was updated through the evaluation-mode spoofing workflow, which applied a fixed spoofed address rather than a fully custom value.
- During the change process, the interface showed the adapter transitioning while the spoofed MAC address field updated to the new value.
- This stage reinforced the same core point as the previous tool: MAC identity can be modified quickly with commodity software, and different tools expose that capability in different ways.

Wireshark Installation and Packet Capture:

- Wireshark 1.12.3 (32-bit) was downloaded from the internal Hacking Tools area, saved locally, and installed on PLABWIN10 together with WinPcap 4.1.3.
- The installation used the default components, added a desktop icon, and completed successfully.
- Once launched, Wireshark was started on the Ethernet interface and immediately began capturing live packets.
- The capture was then stopped to permit analysis rather than continuous collection.
- Using Edit > Find Packet with the Filter option, traffic associated with IP address 192.168.0.1 was highlighted in the packet list.
- The filtered results were reviewed and then saved as a capture file named PLAB on the desktop.
- This stage demonstrated both packet acquisition and post-capture filtering workflow inside a widely used network analysis tool.

SNIFFING ANALYSIS AND PREVENTION

Sniff-O-Matic Capture and Search Workflow:

- Sniff-O-Matic was downloaded from the internal Hacking Tools page and installed on PLABWIN10 with default options.
- During setup, the installer prompted to disable User Account Control, after which the system restarted and required reconnecting to the lab device.
- After restart, the program was opened from the Start menu and run in evaluation mode.
- Packet capture was started from within Sniff-O-Matic, after which Internet Explorer was opened to generate traffic against the intranet site.
- Returning to Sniff-O-Matic, the Options > Find feature was used to search for the string "intranet" in captured data.
- The matching packet was identified with the binoculars marker, and the corresponding details were reviewed in the hexadecimal pane.
- This workflow showed how application data or site references can be located inside sniffed traffic once packets have been captured.

Sniffing Prevention and XArp Utility:

- The prevention portion of the module emphasized protocol hardening and network control measures including HTTPS over HTTP, SFTP over FTP, switching over hubs, DHCP Snooping, Dynamic ARP Inspection, Source Guard, encryption, and anti-sniffing detection tools.
- XArp 2.2.2 was downloaded and installed on PLABWIN10 from the internal Hacking Tools page.
- If WinPcap was not already present, the XArp workflow also installed WinPcap 4.1.2 as a dependency.
- After launch, XArp opened with the default security level set to basic.
- The security level was then raised to aggressive, after which the application reported ARP-based attack detection activity in the monitoring pane.
- The XArp logfile was opened to review the event output in more detail.
- This stage connected packet sniffing awareness to ARP-spoofing detection and showed how protective tooling can be layered alongside safer protocol and switching practices.

Key Commands, Tools, and Artifacts by System:

PLABWIN10 - MAC Spoofing / Packet Capture / Prevention Tooling
Change MAC Address installed and used with multiple Fill modes
SMAC 2.0 installed and evaluation-mode MAC spoofing applied
Wireshark-win32-1.12.3.exe installed with WinPcap 4.1.3
Wireshark capture started on Ethernet; filter applied for 192.168.0.1
Capture saved locally as PLAB
snifftrial.exe installed and Sniff-O-Matic capture started
String search performed for "intranet" inside captured traffic
xarp-2.2.2-win.exe installed and security level raised to aggressive
XArp logfile opened and reviewed
Primary files and artifacts observed
Downloaded installers for Change MAC Address, SMAC 2.0, Wireshark, Sniff-O-Matic, XArp, WinPcap components, and the saved Wireshark capture file named PLAB.

DEFENSIVE MAPPING AND SUMMARY

Technique Summary:

- The completed work covered MAC address manipulation, MAC spoofing, packet capture, traffic filtering, packet searching, and ARP-based attack detection.
- The recurring theme across the module was that visibility at both the interface layer and the packet level can be altered, observed, and analyzed with widely available tooling.
- The prevention side of the work showed that protocol choice, switched infrastructure, encryption, and ARP-monitoring controls all matter when reducing exposure to sniffing activity.

MITRE ATT&CK Mapping:

- Collection: T1040 Network Sniffing - packet capture and inspection with Wireshark and Sniff-O-Matic aligned directly with passive collection of traffic in transit.
- Credential Access / Collection context: T1557 Adversary-in-the-Middle - sniffing risk in this module was closely tied to traffic interception and relay concepts on local networks.
- Credential Access / Collection context: T1557.002 ARP Cache Poisoning - the XArp portion mapped most closely to detection and prevention of ARP-based interception activity.
- Defense Evasion context: T1036 Masquerading - MAC spoofing with Change MAC Address and SMAC most closely aligned with identity manipulation to appear as a different device.
- Collection / Exposure context: weak use of HTTP and FTP in the module reinforced how sniffing becomes more valuable to an adversary when traffic is transmitted over insecure protocols.

Detection Opportunities:

- Unexpected MAC changes, shifting manufacturer identifiers, or unusual adapter values can be signs of spoofing or evasion attempts.
- Packet capture tools operating on endpoints may be detectable through software installation, driver changes, capture file creation, and abnormal interface access patterns.
- Searchable packet content illustrates how weak protocols can leak meaningful strings, hostnames, or identifiers into captured traffic.
- ARP monitoring and log review provide useful signals for identifying local spoofing or poisoning activity in switched environments.

Defensive Controls:

- Prefer HTTPS over HTTP and SFTP over FTP to reduce exposure of sensitive traffic to passive inspection.
- Use switches instead of hubs to reduce trivial packet visibility across shared segments.
- Deploy DHCP Snooping, Dynamic ARP Inspection, and Source Guard where supported to reduce ARP-based manipulation risk.
- Use endpoint monitoring, software control, and capture-tool awareness to detect unauthorized sniffing utilities.
- Layer protective monitoring with utilities such as XArp and with broader network encryption practices.

Completion Verification:

- Work was completed across MAC address changing, SMAC-based spoofing, Wireshark installation and capture, Sniff-O-Matic capture and search, and XArp-based prevention review.
- Completion was evidenced by visible MAC changes, spoofed adapter values, successful packet capture on Ethernet, filtered Wireshark results, saved capture output, located Sniff-O-Matic search results, and XArp alert/log output at higher detection sensitivity.
- The report therefore documents the full workflow from interface identity manipulation through packet-level inspection and prevention-oriented monitoring.

Key Findings:

- 1) MAC identity can be changed quickly with common tools, which affects trust and attribution at the local network layer.
- 2) Wireshark and Sniff-O-Matic both provide useful traffic visibility, but they differ in how they present capture and search workflows.
- 3) Weak protocols can expose meaningful traffic content that becomes easy to locate once packets are captured.
- 4) ARP-focused monitoring tools add an important defensive layer against local spoofing-based sniffing scenarios.
- 5) The most valuable defensive takeaway was that prevention is not a single control but a combination of protocol hardening, network architecture, monitoring, and encryption.

Issues / Deviations / Notes:

- The module was most useful when the tools were compared rather than treated in isolation, since each one emphasized a different part of the sniffing problem.
- MAC spoofing stages highlighted identity manipulation at the adapter level, while Wireshark and Sniff-O-Matic highlighted capture and content inspection at the packet level.
- The prevention portion was stronger once tied back to the earlier capture stages, because XArp and the anti-sniffing guidance made more sense after traffic and spoofing behavior had been observed.

Reflection:

- This work tied together interface-layer identity, packet capture, traffic search, and ARP-based prevention in a way that made sniffing both easier to understand and easier to defend against.
- The strongest takeaway was that packet visibility is only one side of the problem; the other side is reducing how much useful data is exposed in transit and how easily local trust can be abused.
- The completed workflow supports future work in ethical hacking and defensive analysis by showing how spoofing, capture, filtering, and ARP monitoring fit together in a practical workflow.